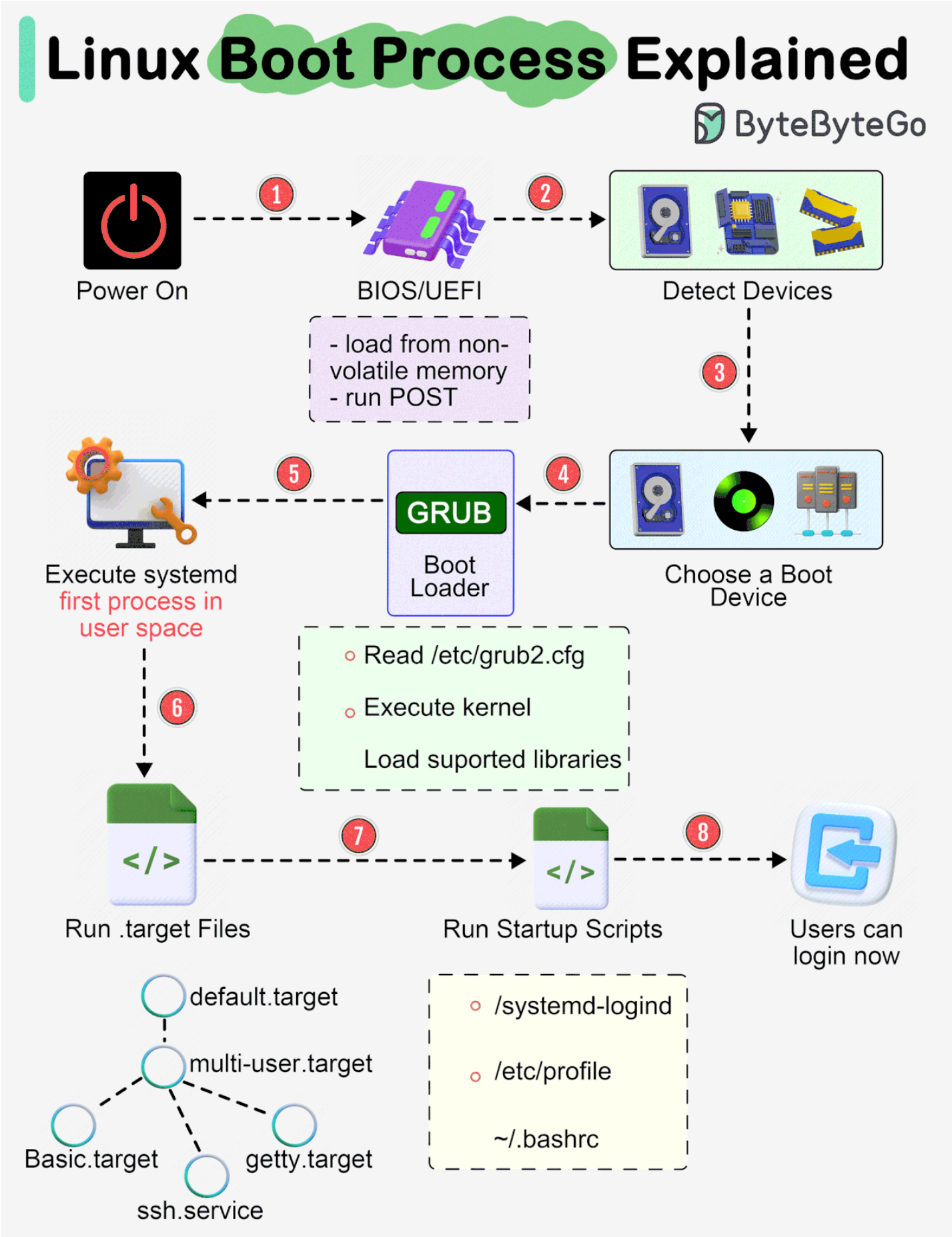
🚀 Linux Booting Process



```
                         GNU GRUB  version 2.04

 *Linux Mint 20 Xfce
  Advanced options for Linux Mint 20 Xfce
  Ubuntu 19.04 (19.04) (on /dev/sda1)
  Advanced options for Ubuntu 19.04 (19.04) (on /dev/sda1)

      Use the ↑ and ↓ keys to select which entry is highlighted.
      Press enter to boot the selected OS, `e' to edit the commands
      before booting or `c' for a command-line.
```

```
         Stopping udev Kernel Device Manager...
[  OK  ] Stopped target Swap.
[  OK  ] Stopped udev Coldplug all Devices.
[  OK  ] Stopped Apply Kernel Variables.
[  OK  ] Stopped target Paths.
[  OK  ] Stopped target Slices.
[  OK  ] Stopped target Sockets.
[  OK  ] Stopped Create Volatile Files and Directories.
[  OK  ] Stopped target Local File Systems.
[  OK  ] Stopped target Local File Systems (Pre).
[  OK  ] Stopped udev Kernel Device Manager.
[  OK  ] Started Cleaning Up and Shutting Down Daemons.
[  OK  ] Stopped Create Static Device Nodes in /dev.
[  OK  ] Stopped Create list of required sta…vice nodes for the current kernel.
[  OK  ] Stopped dracut pre-udev hook.
[  OK  ] Stopped dracut cmdline hook.
[  OK  ] Closed udev Kernel Socket.
[  OK  ] Closed udev Control Socket.
         Starting Cleanup udevd DB...
[  OK  ] Started Cleanup udevd DB.
[  OK  ] Started Setup Virtual Console.
[  OK  ] Reached target Switch Root.
[  OK  ] Started Plymouth switch root service.
         Starting Switch Root...
```

🧩 1. BIOS / UEFI

👉 **BIOS = Basic Input Output System**
👉 **UEFI = Unified Extensible Firmware Interface**

- This is the **first program** that runs when you power on the system
- It performs **POST (Power-On Self Test)**
- It checks hardware like:
  - RAM
  - CPU
  - Keyboard
  - Storage devices

👉 After checking, it finds the **bootable device** (HDD/SSD/USB)

👉 Simple:
Checks hardware and finds where OS is stored

💽 2. MBR / GPT

👉 **MBR = Master Boot Record**
👉 **GPT = GUID Partition Table**
👉 **GUID = Globally Unique Identifier**

- This is the **first part of the disk**
- It contains:
  - Partition information
  - Bootloader location

👉 Difference:

- MBR → old system (limited partitions)
- GPT → modern system (supports large disks)

👉 Simple:
Tells where bootloader is located

⚙️ 3. GRUB

👉 **GRUB = GRand Unified Bootloader**

- It loads the operating system
- Shows a menu (if multiple OS are installed)
- Lets you choose kernel

👉 Steps:

1. Reads configuration file
2. Shows menu
3. Loads kernel

📂 File:

```
/boot/grub2/grub.cfg
```

👉 Simple:
Loads Linux operating system

🧠 4. Kernel

👉 **Kernel = Core of Operating System**

- Loaded into RAM
- Starts controlling:
  - CPU
  - Memory
  - Devices

👉 Responsibilities:

- Process management
- Memory management
- Device drivers

👉 Simple:
Brain of the system

💾 5. initramfs

👉 **initramfs = Initial RAM File System**
👉 Old: **initrd = Initial RAM Disk**

- Temporary file system in RAM
- Used before real system starts

👉 Work:

- Loads required drivers
- Prepares system

👉 Simple:
Temporary setup for booting

🔄 6. init / systemd

👉 **init = Initialization process**
👉 **systemd = System Daemon**

- First process started (**PID = 1**)
- Starts all services like:

- Network
- SSH
- Background services

👉 Modern Linux uses **systemd**

👉 Simple:
Starts and manages system services

🎯 7. Runlevels / Targets

👉 **Runlevel = System mode**
Runlevel decides how the system works after boot

Runlevel	Meaning
0	Shutdown (system off)
1	Single-user mode (maintenance)
2	Multi-user (no network, rarely used)
3	Multi-user (CLI with network)
4	Not used (custom)
5	Graphical mode (GUI)
6	Reboot

👉 systemd uses:

- `multi-user.target`
- `graphical.target`

👉 Simple:
Defines how system runs (CLI or GUI)

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8. Login Stage

- System shows login screen
- User enters username and password

👉 Now system is ready to use ✅

🔥 Complete Boot Flow

👉

BIOS (Basic Input Output System)

➡️ MBR (Master Boot Record) / GPT (GUID Partition Table)

➡️ GRUB (GRand Unified Bootloader)

➡️ Kernel (Core of OS)

➡️ initramfs (Initial RAM File System)

➡️ systemd (System Daemon)

➡️ Login